

BOLT: Autonomous Driverless Shuttle

Technical Overview and Achievements

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Outline

- 1 Introduction
- 2 Mechatronics Systems
- 3 Electrical Systems
- 4 Software Systems
- 5 Achievements & Future Goals



Figure: BOLT

BOLT Driverless Shuttle

- Autonomous vehicle for campus transportation
- Multi-sensor perception system
- Advanced control algorithms
- Progressive milestone achievements



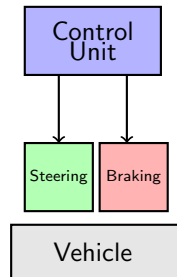
Figure: BOLT

What is Drive-By-Wire?

Electronic control of vehicle functions replacing mechanical linkages

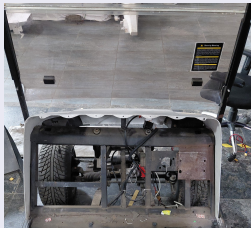
Key Advantages:

- Eliminates mechanical connections
- Enables autonomous control
- Precise electronic actuation
- Enhanced safety through redundancy
- Real-time control feedback



Foundation for autonomous vehicle dynamics

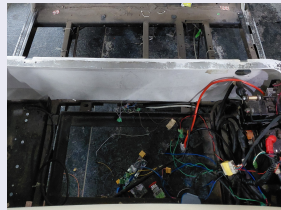
Steering Control



Features:

- Linear actuator-based
- Precise angular control
- Electronic steering commands
- Fast response time

Braking System



Features:

- Linear actuator integration
- Variable braking force
- Safety-critical redundancy
- Emergency stop capability

Power Distribution

48V Battery System

- Powers motor controller
- High current delivery
- Primary propulsion power

24V Battery System

- Feeds inverter
- Electronics (PCBs, Controllers, etc.)
- Compute unit

Control Electronics

Custom PCBs

- Throttle control interface
- Brake control interface
- Real-time response

PID Control Loop

- Wheel encoder feedback
- Speed regulation
- Precise vehicle control

Multi-Modal Sensor Suite

Vision Systems

3 × ZED Stereo Cameras

- Lane detection
- Ground plane estimation
- Short-range obstacles (YOLO)

RADAR

- All-weather detection
- Velocity measurements

LiDAR Systems

2D LiDAR

- Front obstacle detection
- High-frequency scanning

3D LiDAR

- 360° environment mapping
- SLAM (Simultaneous Localization and Mapping)
- Point cloud generation

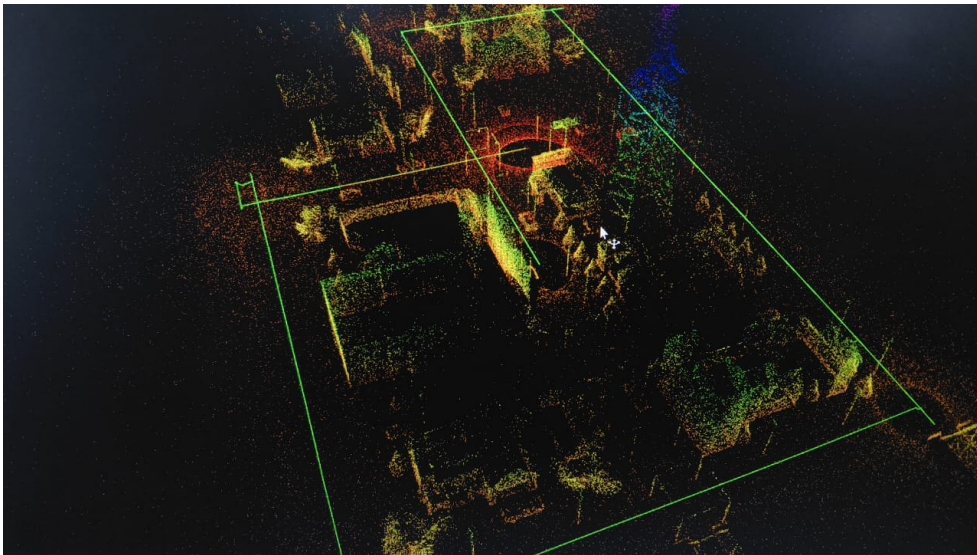
Sensor Fusion

- **Extended Kalman Filter (EKF)** fuses all odometry sources
- Provides robust pose estimation
- Combines:
 - Wheel odometry
 - Visual odometry (cameras)
 - LiDAR odometry
 - IMU data

SLAM with 3D LiDAR

- Real-time map building
- Continuous localization
- Environment understanding

Fast-LIO Map



Path Planning

Occupancy Grid Mapping

Generation Process:

- Created from obstacle detection data
- Probabilistic representation of environment
- Grid cells marked as occupied/free/unknown
- Real-time updates from sensors

A* Path Planner

Planning Algorithm:

- Optimal path computation
- Heuristic-based search (A*)
- Obstacle avoidance
- Efficient route generation

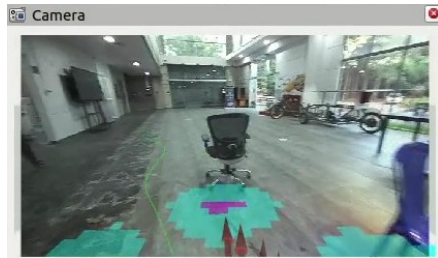


Figure: Path Planning

Trajectory Control

Controller Options

Multiple controllers have been implemented: **1.**

Pure Pursuit

- Geometric path following
- Look-ahead distance based

2. MPPI (Model Predictive Path Integral)

- Considers vehicle dynamics
- Optimal control trajectory

3. DWB (Dynamic Window Approach)

- Velocity space search
- Dynamic constraints handling

Trajectory Control

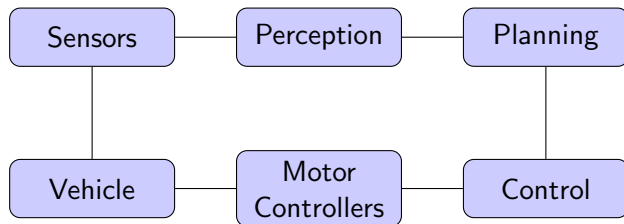
Multiple controllers can be used:

- **Pure Pursuit:** Path following
- **MPPI:** Model Predictive Path Integral
- **DWB:** Dynamic Window Approach

Provides:

- Longitudinal control (speed)
- Lateral control (steering)

Closed-Loop Control System



- Continuous feedback loop ensures safe operation
- Real-time sensor measurements guide control decisions
- Motor controllers execute planned trajectories

Milestones Achieved

2023: Foundation

- **500 meter autonomous run** completed successfully
- Demonstrated basic autonomous driving capability
- No obstacle avoidance required

2024: Advanced Navigation

- **Obstacle detection and avoidance** showcased
- Real-world navigation in dynamic environments
- Multi-sensor fusion validated



Future Roadmap

Next Milestone: Extended Range

- **2 km autonomous run:** Gagendra Circle → Main Gate
- Complex route with multiple scenarios
- Real campus traffic conditions

Ultimate Goal: Campus Shuttle Service

- Autonomous shuttle service throughout campus
- Scheduled routes and stops
- Safe, reliable transportation for students and staff
- Contribution to sustainable campus mobility

Building the future of autonomous transportation, one milestone at a time

Technical Excellence

- Robust mechanical systems
- Reliable electrical architecture
- Advanced software stack
- Multi-sensor perception

Proven Progress

- Successful demonstrations
- Progressive complexity
- Clear future vision
- Practical application